

Frames and matrix designs

Counting real Hadamard matrices

Difficulty: extreme **Importance:** interesting to the community**Status:** Solved**Topic:** Enumeration of orthogonal sign matrices

Rating rationale: Known general upper bounds still have a quadratic exponent, while the conjecture asks for an exponent of order $n \log n$. The question concerns the abundance of exact flat orthogonal transforms, with connections to structured matrix constructions and elimination.

Negative resolution - 2026-09-12

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The counting conjecture is false. The [complete proof](#), Lemma 1 and Theorem 1, constructs an injection from two labeled order- m Hadamard matrices and a perfect matching of $2m$ row labels. It gives

$$H(2m) \geq (2m-1)!! H(m)^2, \quad H(2^k) \geq 2^{2^k(k-1)(k-2)/8} \quad (k \geq 2).$$

The resulting exponent has order $n(\log n)^2$ along powers of two, contradicting every bound $2^{Cn \log_2 n}$ with fixed C . [Proof PDF](#) · [Standalone proof TeX](#).

The full argument passed a separate [independent Codex-agent mathematical and source-scope review](#). The [submission record](#) preserves the supplied source, exact checks, document conversion and bounded public fork/branch/PR and literature search. Substantial AI assistance is disclosed; this is informal automated review, not external human peer review or formal verification.

Ferber, Jain and Zhao retain attribution for the conjecture and prior upper bound. This result does not settle existence at every admissible order or determine a matching upper bound for the count. The original statement and historical context below are retained; the difficulty and importance ratings are historical.

Statement

For each positive integer n , let

$$H(n) = \#\{A \in \{-1, 1\}^{n \times n} : AA^T = nI_n\}.$$

Conjecture. There is an absolute constant $C > 0$ such that

$$H(n) \leq 2^{Cn \log_2 n}$$

for every positive integer n divisible by four.

The count is of individual matrices with their row and column labels. Matrices related by signed permutations are not identified. The statement does not require that a Hadamard matrix exist at every such order.

Known bounds and numerical significance

Ferber, Jain and Zhao prove that some absolute $c_H > 0$ gives $H(n) \leq 2^{(1-c_H)n^2/2}$ for every sufficiently large multiple of four. Whenever $H(n) > 0$, distinct row permutations of one Hadamard matrix give $H(n) \geq n!$. Thus the conjectured exponent has the smallest possible order along orders admitting such matrices.

Dividing a Hadamard matrix by $\sqrt{n}$ produces an orthogonal transformation whose entries all have the same magnitude. The enumeration asks how many exact sign designs can underlie these transforms. Peca-Medlin's work connects Hadamard enumeration for butterfly constructions to structured orthogonal matrices and Gaussian elimination. This is distinct from the Hadamard existence conjecture and from bounds on elimination growth for a given matrix.

References and status check

- A. Ferber, V. Jain and Y. Zhao, *On the number of Hadamard matrices via anti-concentration*, *Combinatorics, Probability and Computing* 31 (2022), 455–477. [DOI](#); [published PDF](#). Conjecture 1.3 on p.456 is the displayed target; Theorem 1.2 gives the upper bound. The [arXiv record](#) also supplies the earlier manuscript, where the conjecture has different numbering.
- [BIRS workshop 24w5204 report \(2024\)](#), Conjecture 28, restates the enumeration target.
- J. Peca-Medlin, *Complete pivoting growth of butterfly matrices and butterfly Hadamard matrices* (2026). [DOI](#), §3 before Proposition 3.1, discusses the general count as conjectural and enumerates specific butterfly constructions.

On 2026-09-11, checked the published statement and later restatements, and searched the exact title, the authors' names, Hadamard enumeration, upper bounds, and 2025–2026 proof/counterexample combinations. No full resolution was located. The butterfly counts apply to restricted families and do not settle this all-matrix upper bound. This is a bounded literature check.